🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD 6.GR.2

Volume with Whole Number Edges

Lesson 10-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes **Level 2 Standard**

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the volume of a rectangular prism with whole-number edges using length × width × height.

Language Objective: I can explain my work using the words volume, rectangular prism, cubic units, and edge.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Volume | **Rectangular prism** | **Cubic units** | **Length, width, height** | **Net** | **Edge**

👆 *Tap any word to see what it means and a picture.*

1. The amount of space inside a three-dimensional solid is its ______________ .

2. A solid with six rectangular faces, like a box, is a ______________ .

3. The unit used to measure volume, like cubic centimeters, is ______________ .

4. The three edge measurements of a rectangular prism are its ______________ .

5. A flat pattern that folds up into a solid is a ______________ .

6. A line segment where two faces of a solid meet is an ______________ .

Reflect — Exit Ticket

A rectangular prism has l = 11 in, w = 5 in, h = 4 in. What is the volume?

A. 220 in^3

B. 55 in^3

C. 220 in^2

D. 200 in^3

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Length, width, height
5. **Notes 5:** Net
6. **Notes 6:** Edge
7. **Watch for this mistake:** *A common mistake in Volume with Whole Number Edges is skipping the key idea: "To find the volume of a box, multiply all three edges: V = length × width × height." — always check your work against this rule before you submit.*
8. **Try It 1:** A. 30 cubic units — *V = length × width × height = $5 \times 3 \times 2 = 30$ cubic units.*
9. **Try It 2:** A. 24 cubic units — *Each layer has 6 cubes and there are 4 layers, so $6 \times 4 = 24$ cubic units. This is the same as length × width × height with one factor being the number of layers.*
10. **Exit Ticket:** A. 220 in^3 — *$V = 11 \times 5 \times 4 = 220$ cubic inches. Remember: volume uses cubic units (in^3).*